



THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFERABAD

DEPARTMENT OF SOFTWARE ENGINEERING

Computer Networks

(Lab no # 02)

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Submitted To:

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Open Cisco Packet Tracer

1. Launch **Cisco Packet Tracer**.
2. A blank workspace will appear.

Add Two PCs

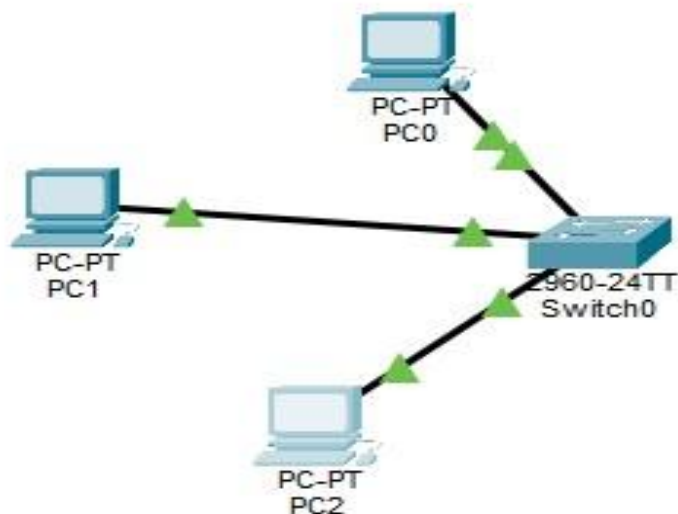
1. From the **bottom left panel**, click **End Devices**.
2. Drag **PC** to the workspace.
3. Drag **another PC** to the workspace.

Now you will see **PC0** and **PC1**.

Connect the PCs

1. Click the **Connections (lightning icon)**.
2. Select **Copper Cross-Over Cable**.
3. Click **PC0 → FastEthernet0**.
4. Click **PC1 → FastEthernet0**.

Now both PCs are physically connected.



Assign IP Addresses

On PC0

1. Click **PC0**.
2. Go to **Desktop** tab.
3. Click **IP Configuration**.
4. Enter:
 - **IP Address:** 192.168.1.1
 - **Subnet Mask:** 255.255.255.0

On PC1

1. Click **PC1**.
2. Go to **Desktop → IP Configuration**.
3. Enter:
 - **IP Address:** 192.168.1.2
 - **Subnet Mask:** 255.255.255.0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::260:70FF:FEAD:822A

Default Gateway:

DNS Server:

○

Verify the Connection (Ping Test)

1. Click **PC0**.
2. Go to **Desktop → Command Prompt**.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::260:70FF:FEAD:822A
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>

```

3. Type:

ping 192.168.1.2

If connected correctly, you will see:

Reply from 192.168.1.2

This means the **connection between both PCs is successful.**

```

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=238ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 238ms, Average = 60ms

C:\>

```

Tracer route:

```
C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1    11 ms    4 ms    3 ms    192.168.1.1

Trace complete.
```

Address Resolution Protocol:

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.2           00d0.d375.15db        dynamic

C:\>
```

Network statistics:

```
C:\>netstat

Active Connections

Proto Local Address          Foreign Address         State
C:\>netstat -n
```

Note: Same process is followed for all the PCs in the setup.